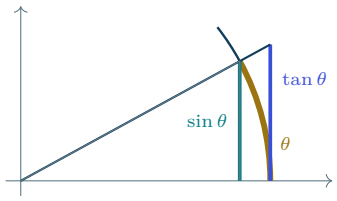
FUNCTIONS LADDER 3 OF 10

Small-angle approximations

For small angles in radians the sine and tangent are close to the angle itself, and the cosine is close to one.

1 / THE RULE AND THE PICTURE

For a small angle in radians $\sin\theta < \theta < \tan\theta$, and all three are nearly equal



$\theta = 0.5$ rad on a unit circle drawn large

In the picture the arc is longer than the vertical side of the small triangle and shorter than the vertical side of the large one, which squeezes

$$\sin\theta \approx \theta, \qquad \tan\theta \approx \theta, \qquad \cos\theta \approx 1 - \tfrac{\theta^2}{2}$$

and the approximations hold only when θ is measured in radians.

2 / HOW GOOD ARE THEY

The errors grow like θ^2 for the ratios, and stay below 1% for $\sin$ up to about 10°

angle	θ (rad)	$\sin\theta$	error of $\sin \approx \theta$	error of $\tan \approx \theta$	error of $\cos \approx 1 - \frac{\theta^2}{2}$
1°	0.0175	0.0175	0.01%	0.01%	3.9×10^{-9}
5°	0.0873	0.0872	0.13%	0.25%	2.4×10^{-6}
10°	0.1745	0.1736	0.51%	1.02%	3.9×10^{-5}
15°	0.2618	0.2588	1.15%	2.30%	2.0×10^{-4}
30°	0.5236	0.5000	4.72%	9.31%	3.1×10^{-3}

At 10° the error in $\sin\theta \approx \theta$ is 0.51% and in $\tan\theta \approx \theta$ twice that, 1.02%; at 30° the sine is still within 4.7% but the tangent is 9.3% out.

3 / WHY / THE NEXT TERM

The next term of each series gives the size of the error

The series are $\sin\theta = \theta - \frac{\theta^3}{6} + \cdots$, $\tan\theta = \theta + \frac{\theta^3}{3} + \cdots$ and $\cos\theta = 1 - \frac{\theta^2}{2} + \frac{\theta^4}{24} - \cdots$. The relative error of $\sin\theta \approx \theta$ is about $\theta^2/6$, which at $10^\circ = 0.1745$ rad gives 0.0051, matching the table, and the relative error of $\tan\theta \approx \theta$ is about $\theta^2/3$, twice as large.
These are the first terms of the Taylor series, and the squeeze $\sin\theta < \theta < \tan\theta$ gives the limit $\sin\theta/\theta \to 1$ as $\theta \to 0$, the fact that makes $\frac{d}{d\theta}\sin\theta = \cos\theta$ true.

4 / THE PENDULUM

The period of a pendulum is nearly independent of amplitude only for small swings

amplitude θ_0	exact T/T_0	$1 + \theta_0^2/16$
5°	1.00048	1.00048
10°	1.00191	1.00190
30°	1.01741	1.01713
60°	1.07318	1.06854

Replacing $\sin\theta$ by θ in the equation of motion gives simple harmonic motion with $T_0 = 2\pi\sqrt{L/g} = 2.006$ s for $L = 1$ m; the correction factor $1 + \theta_0^2/16$ is accurate to a few parts in a thousand up to 30°.

5 / USES

Angular size, Snell's law and apparent depth

The Moon is 3474 km across at $384\,400$ km, so its angle is $\theta \approx \dfrac{3474}{384\,400} = 0.00904$ rad $= 0.518^\circ$, and one arcsecond is 4.85×10^{-6} rad.
For light crossing a boundary Snell's law $n_1 \sin\theta_1 = n_2 \sin\theta_2$ becomes $n_1\theta_1 \approx n_2\theta_2$ near the normal, so a pool of real depth 3 m and $n = 1.33$ looks about $3/1.33 = 2.26$ m deep when viewed from above.

ASIDE / DEGREES BREAK IT

Using degrees in $\sin\theta \approx \theta$ is the classic mistake, since $\sin 1^\circ = 0.01745$ and not 1.

LINKS TO OTHER SHEETS

The exact pendulum period uses *Hypergeometric functions*; the previous sheet is *Hyperbolic functions* and the next is *Exponential functions and the number e*.

Convention. θ in radians; relative error is (approximation − exact)/exact in absolute value; $g = 9.81$ m/s^2.